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IT4010 – Research Project - 2024

Topic Assessment Form

Project ID : R25-020

1. Topic (12 words max)

Leveraging Data Analytics and Behavioral Analysis for Improved Testimony and Incident Correlation

2. Research group the project belongs to

CEAI - Centre of Excellence in AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The research problem addresses the significant limitations and inefficiencies in traditional criminal investigation processes, which often rely heavily on manual efforts and subjective human interpretation. Investigators are responsible for collecting, organizing, and analyzing large volumes of case data, such as testimonies, incident reports, and evidence. This manual process is prone to human error, including misinterpretations, biases, inconsistencies in judgment, and fatigue, all of which can lead to inaccuracies, overlooked details, and delayed case resolution.

In addition, the process of cross-referencing testimonies and correlating new incidents with historical cases is slow, fragmented, and manual. This inefficiency makes it difficult to identify patterns, connections, or inconsistencies across cases, often resulting in missed opportunities for timely case resolution. The reliance on human judgment and manual analysis, coupled with limited real-time support during interrogations, hinders the effectiveness and accuracy of criminal investigations.

Currently, very few systems are available globally to assist in such investigations, and the few that exist have very limited capabilities. Most existing systems focus on individual tasks, such as lie detection or basic data management, and lack the integration of multiple advanced technologies that are needed for comprehensive support.

This combination of human error, inefficiency, and the lack of advanced, integrated systems in current practices underscores the need for a more robust, AI-driven solution to enhance the accuracy, efficiency, and effectiveness of criminal investigations.

References:

1. A. T. A. Sadiqur Rahman and A. Kaur, "The Application of Artificial Intelligence in the Administration of Criminal Law and the Procedures Involved in Criminal Cases," *2024 International Conference on Advances in Computing Research on Science Engineering and Technology (ACROSET)*, Indore, India, 2024, pp. 1-5. doi: 10.1109/ACROSET62108.2024.10743336.

2. V. Singh, A. Dixit, S. Pandey, B. V. Kumar, V. Pachouri, and M. Sahu, "Role of Artificial Intelligence in Criminal Investigation," *2023 International Conference on Quantum Technologies, Communications, Computing, Hardware and Embedded Systems Security (iQ-CCHESS)*, Kottayam, India, 2023, pp. 1-4. doi: 10.1109/iQ-CCHESS56596.2023.10391288.

3. Baltrūnienė, J. (2023). Place of artificial intelligence in the detection and investigation of crime: the present state and future perspectives. Current Problems of Forensic Science, 26, 43–58. doi: 10.52097/pwk.5431.

4. I. Lakshan, L. Wickramasinghe, S. Disala, S. Chandrasegar and P. S. Haddela, "Real Time Deception Detection for Criminal Investigation," 2019 National Information Technology Conference (NITC), Colombo, Sri Lanka, 2019, pp. 90-96, doi: 10.1109/NITC48475.2019.9114422.

5. AI-driven Justice: Evaluating the Impact of Artificial Intelligence on Legal Systems - Rachid Ejjami - IJFMR Volume 6, Issue 3, May-June 2024. doi: 10.36948/ijfmr.2024.v06i03.23969.

6. S. S. Das, S. Patnaik, P. Pattnayak and A. Mohanty, "An Advancement in Forensic and Criminal Investigation Through Artificial Intelligence," 2023 OITS International Conference on Information Technology (OCIT), Raipur, India, 2023, pp. 920-926, doi: 10.1109/OCIT59427.2023.10431257.

7. Jasleen Kaur Sabherwal, Kirandeep Kaur, "The Role of Artificial Intelligence in Improving Criminal Justice System", International Journal of Science and Research (IJSR), Volume 13 Issue 4, April 2024, pp. 999-1002, doi: 10.21275/SR24412153835.

8. Y. Hu, Y. Li, T. Yang and Q. Pan, "Short Text Classification with A Convolutional Neural Networks Based Method," 2018 15th International Conference on Control, Automation, Robotics and Vision (ICARCV), Singapore, 2018, pp. 1432-1435, doi: 10.1109/ICARCV.2018.8581332.

9. S. Lai, L. Xu, K. Liu, and J. Zhao, “Recurrent Convolutional Neural Networks for Text Classification”, AAAI, vol. 29, no. 1, Feb. 2015.

10. Alexis Conneau, Holger Schwenk, Loïc Barrault, and Yann Lecun. 2017. Very Deep Convolutional Networks for Text Classification. In Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers, pages 1107–1116, Valencia, Spain. Association for Computational Linguistics.

11. Higginson, Angela & Eggins, Elizabeth & Mazerolle, Lorraine. (2017). Police techniques for investigating serious violent crime: A systematic review. Trends and Issues in Crime and Criminal Justice. 539.

12. C. Rigano, "Using Artificial Intelligence to Address Criminal Justice Needs," *National Institute of Justice (NIJ)*, Oct. 8, 2018. [Online]. Available: https://nij.ojp.gov/topics/articles/using-artificial-intelligence-address-criminal-justice-needs. [Accessed: Dec. 4, 2024].
13. "Investigation and Evidence Gathering," *Criminal Justice Research*, [Online]. Available: https://criminal-justice.iresearchnet.com/criminal-justice-process/investigation-and-evidence-gathering. [Accessed: Dec. 4, 2024].
14. EclipseForensics, "Comparing Traditional vs. Modern Forensic Methods: A Brief Guide," EclipseForensics, Oct. 2, 2024. [Online]. Available: https://eclipseforensics.com/comparing-traditional-vs-modern-forensic-methods-a-brief-guide. [Accessed: Dec. 4, 2024].
15. Sri Lanka Law, Code of Criminal Procedure Act, [Online]. Available: https://www.srilankalaw.lk/c/217-code-of-criminal-procedure-act.html. [Accessed: Dec. 5, 2024].

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed AI-driven solution enhances criminal investigations through a multi-modal approach that integrates advanced technologies for data collection, testimony evaluation, and incident correlation. The system processes live video feeds, heart rate monitoring, and audio data from suspects during interviews to analyze behavioral cues and detect signs of stress or deception.

Key components of the system include:

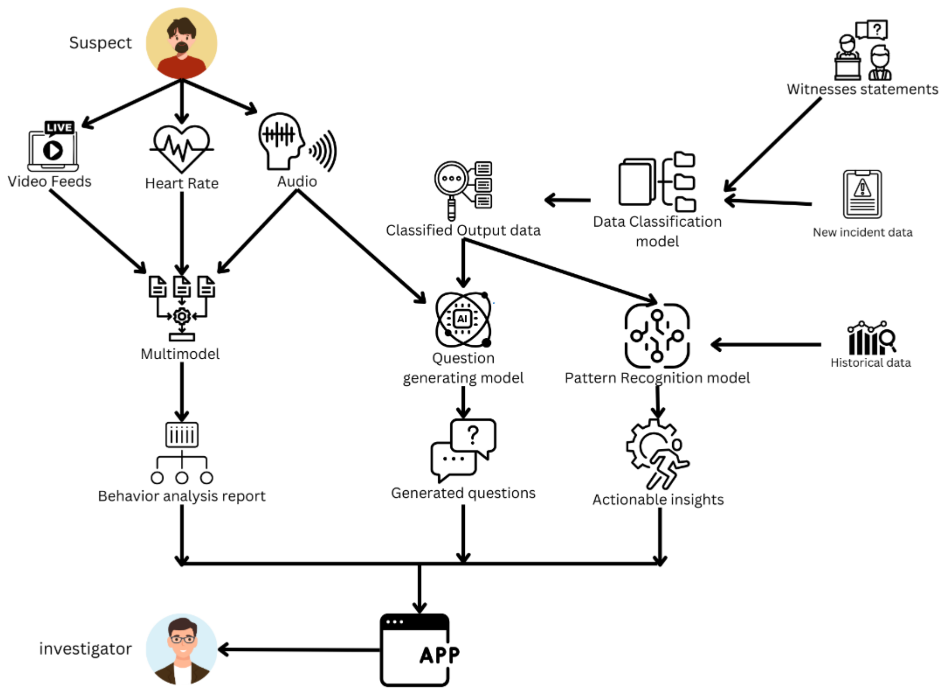
1. Automated Data Collection and Management: This component uses OCR and NLP to automatically extract and organize relevant information from police reports, witness statements, and other documents, reducing human error in data handling.

2. Behavioral and Emotional Analysis: The system analyzes video feeds, heart rate monitoring, and audio data to detect emotional and physiological cues, such as stress or discomfort, providing real-time feedback on the credibility of testimonies.

3. Dynamic Question Generation and Response Analysis: NLP techniques are used to analyze responses during interrogations, generating contextually relevant follow-up questions in real-time, enhancing the quality and depth of interviews.

4. Incident Correlation and Pattern Identification: Machine learning algorithms compare new incidents with historical case data to identify patterns and similarities, helping investigators uncover critical connections and insights.

These outputs are delivered to investigators through an intuitive app, streamlining the investigation process and reducing human error. The integrated system offers comprehensive, real-time support, enabling more efficient, accurate, and timely criminal investigations by automating key tasks and providing actionable insights.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Expertise Requirements

1. **Criminal Justice and Forensic Knowledge**
   A deep understanding of criminal law, investigation procedures, and forensic science is critical to ensure the system aligns with legal standards. This knowledge base will guide the system's design to accurately interpret emotional analysis and testimony evaluation within the legal framework.
2. **AI and Machine Learning Expertise**
   Advanced knowledge of AI techniques, including Natural Language Processing (NLP), machine learning (ML), and computer vision, is essential. This encompasses the development of algorithms capable of analyzing witness statements, physiological cues, and contextual data using rigorously preprocessed information.
3. **Behavioral Psychology and Physiological Understanding**
   Comprehensive knowledge of behavioral psychology and physiology is necessary to accurately assess suspect behavior. This includes understanding the links between physiological responses, such as heart rate fluctuations, and emotional states like stress or deception, which can be evaluated through video and physiological monitoring.

Data Requirements

The system will rely on diverse datasets, including historical case data, witness statements, audio/video recordings, heart rate data, and incident reports. These datasets will undergo rigorous preprocessing to ensure quality and accuracy. Preprocessing steps include:

- Cleaning and filtering data to remove inconsistencies.
- Anonymizing sensitive data to comply with privacy laws.
- Standardizing formats to ensure compatibility with AI algorithms.
- Enhancing data quality through augmentation and feature extraction.

Data Sources

The system will use the following datasets as primary references for development and training, while additional necessary data will be generated using generative AI to complement and enhance the dataset quality:

1. Speech Emotion Recognition (English) Dataset
2. Speech Emotion Recognition Kaggle Project
3. Cohn-Kanade (CK) Dataset
4. FER-2013 Facial Expression Dataset
5. AffectNet Dataset for Facial Expressions
6. ASCERTAIN Dataset (Emotion Recognition)
7. Crime Data (U.S. Government Catalog)

8. Objectives and Novelty

Main Objective			
Develop an AI-driven system that enhances the efficiency and accuracy of criminal investigations. By automating data collection, analyzing testimonies in real-time, and identifying patterns across incidents, the system reduces human error and provides investigators with actionable insights. The integration of video, heart rate, audio, and witness data, along with machine learning, aims to streamline investigative processes and improve decision-making, accelerating case resolution.			
Member Name	Sub Objective	Tasks	Novelty
Karunarathna K A D	Automate the extraction, organization, and management of case data using AI technologies	- Implement Optical Character Recognition (OCR) and Natural Language Processing (NLP) techniques to automate the extraction of key information from police reports, witness statements, and other relevant documents. - Develop APIs to integrate the system with existing law enforcement databases, ensuring real-time synchronization and access to critical case data. - Build a secure, relational or NoSQL database system to store and manage both structured (e.g., suspect data, case details) and unstructured (e.g., witness testimonies, images) data. - Create an intuitive, searchable dashboard that enables investigators to quickly filter and retrieve information based on various criteria (e.g., date, location, involved individuals).	The novelty lies in automating the extraction, organization, and management of data from multiple sources in real-time, significantly reducing human error and streamlining the data access process. This integration ensures faster, more accurate decision-making during investigations by making all relevant data instantly available to investigators.

Abeywardana B A I S	Develop a multi-modal system for real-time analysis of emotional and physiological behavior during suspect interrogations	- Develop a facial emotion detection system using computer vision techniques, employing convolutional neural networks (CNNs) to analyze video footage from interviews and identify emotions such as fear, anger, or stress. - Integrate heart rate monitoring with other behavioral analysis tools to detect physiological responses that may indicate deception or emotional distress during interrogations. - Implement audio analysis algorithms to assess voice patterns, such as changes in tone, pitch, and speed, to detect signs of nervousness or dishonesty. - Provide real-time feedback to investigators through the system's interface, highlighting critical emotional or vocal irregularities during testimony evaluation.	The unique aspect of this system is its multi-modal approach, combining facial emotion recognition, heart rate data, and audio analysis into a unified tool. This allows for a comprehensive assessment of a suspect's emotional and physiological responses, offering real-time insights into their behavior that were not previously available with traditional investigative methods.
Ediriweera E R L C	Implement a dynamic, real-time question generation and response analysis system to enhance interrogation efficiency	- Design an NLP-based algorithm that continuously analyzes the suspect's responses in real-time to evaluate coherence, relevance, and consistency. - Develop a dynamic questioning module that generates follow-up questions based on the analysis of the suspect's answers, aiming to probe areas of inconsistency or suspicion.	The innovation in this component lies in its ability to generate contextually relevant, follow-up questions in real-time, based on dynamic analysis of responses. This significantly enhances the quality of interviews by ensuring that questioning adapts to the evolving

		- Implement a feedback loop where investigators can rate the effectiveness of the generated questions, which will be used to refine the model's question generation for future interrogations. - Create an AI-driven strategy that adjusts the questioning approach depending on emotional cues and identified contradictions in the testimony.	conversation, addressing inconsistencies and probing deeper into potentially deceptive statements.
Fernando N D R N	Leverage machine learning algorithms to identify patterns, correlations, and discrepancies between new incidents and historical case data	- Utilize machine learning algorithms to analyze and compare new incidents with historical case data, identifying patterns based on crime type, geographic location, and individuals involved. - Develop clustering and classification models to correlate new incidents with similar past cases, flagging them as potential leads. - Implement semantic analysis to compare testimonies from different witnesses, highlighting discrepancies or matching statements across individuals. - Build an incident alert system that notifies investigators of significant patterns or anomalies detected in the data, enabling them to focus on the most relevant connections.	The novel aspect of this component is its predictive capability. By proactively suggesting similar past incidents and detecting potential patterns between new and historical cases, the system offers a layer of intelligence that accelerates investigations and helps investigators uncover connections that might not have been apparent through manual comparison.

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Topic Assessment Form

9. **Supervisor details**

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Lakmini	Abeywardhana	
Co-Supervisor	Ms.	Pipuni	Wijesiri	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

They are dealing with high sensitive data. but those will be use anonymously. Without hiddering Privacy.
They wil be taking ethical clearance ~~data~~.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	**Signature**

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.